

Divyansh Tulsyan

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EDUCATION

The University of Hong Kong – Year 4

09/2023 – 06/2027

- Bachelor of Engineering, Major: Computer Science (CGPA: 3.44/4.3)
- Chairperson (HKU Student's Cricket Club), Table Tennis Team (HKU), Dance Team (Astreas)
- HKU Undergraduate Entrance Scholarship for Outstanding Academic Talents

The University of New South Wales – Sydney, Australia

01/2026 – 05/2026

- Exchange Semester, courses include Computer Networks, Algorithm Design and Analysis, and Artificial Intelligence

EXPERIENCE

System Analyst Intern – Aereve Company Ltd, Hong Kong

07/2026 – 09/2026

- Optimized an AI-powered eKYC screening pipeline using Python, Flask, SQLite, SerpAPI and DeepSeek, reducing end-to-end latency by ~80% through concurrency, caching and inference optimization while improving classification accuracy to >97%.
- Designed an entity-clustering algorithm to distinguish individuals and organizations with similar names using extracted identity attributes, weighted similarity scoring, contradiction checks, and confidence thresholds.
- Built an MCP-based agentic workflow to automate multi-step eKYC investigations, orchestrating web search, evidence extraction, entity clustering, and analyst review with traceable tool calls and human validation.

Software Developer Intern – Decoder, Hong Kong

06/2025 – 08/2025

- Engineered a scalable booking module into a CRM platform using Next.js, Node.js, TypeScript, and MongoDB, centralizing service catalogs, employee schedules, and reservations to reduce operational overhead. Automated the reservation workflow by building booking flows in WhatsApp Manager and integrating them via webhooks.
- Designed a role-based authentication and authorization system with granular permissions, secure tenant-level data isolation and dynamic interface access.
- Developed seven web-scraping pipelines using multiprocessing and built supporting APIs, reducing data aggregation time by 70% and supplying structured data to the company's chatbot knowledge base.

PROJECTS

Low-Latency Electronic Trading Engine – C++, Linux

07/2026 – 07/2026

- Built a C++ electronic trading stack with UDP binary market-data feeds, sequence-gap detection, a price-time-priority matching engine, and TCP order entry supporting acknowledgements, fills, cancellations, and rejects.
- Engineered an autonomous market-making client with order lifecycle and position management, then instrumented and optimized latency-critical paths using percentile measurements, allocation-free serialization, CPU-affinity experiments, sanitizers, and Linux perf.

Unix Shell & Process Runtime – C++, Linux

07/2025 – 07/2025

- Built a C++ Unix shell/runtime with a custom lexer/parser, pipelines, I/O redirection, conditional execution, POSIX process management, and RAII-based file-descriptor handling.
- Implemented job control, signals, /proc inspection, resource limits, Linux namespaces, and ptrace tracing, with ASan/UBSan, stress testing, and fuzzing for reliability.

Multiplayer Big Two Game – Java

10/2024 – 11/2024

- Developed a multiplayer "Big Two" card game in Java with full rule implementation and a user-friendly GUI.
- Utilized multi-threading to support real-time gameplay for multiple clients, ensuring smooth interactions.
- Implemented game logic for card validation, turn management, and win conditions.

COMPETITIONS

HKU Pitch New Tech Ideas (Winner) – VibeUI.

03/2026 – 03/2026

- Working on VibeUI, a startup focused on bringing AI into the graphical era, recently awarded up to HK\$100,000 in funding support after winning HKU's Pitch New Tech Ideas 2026.

UBS Coding Challenge (Top 5) – Python

09/2025 – 09/2025

- Designed and implemented algorithmic solutions using graphs, networks, cybersecurity, computer vision, and API integrations to solve complex computational problems.

SKILLS

- **Programming Languages:** C++, Python, Java, C, SQL, JavaScript, TypeScript, R, HTML, CSS
- **Frameworks:** Express.js, Next.js, React.js, Node.js, Flask
- **Databases:** MySQL, MongoDB, SQLite
- **Tools and Technologies:** Linux, Docker, Git, GitHub, Codex, Claude Code
- **Areas of Experience:** Algorithms & Data Structures, Systems Programming, Networking, Performance Engineering, Database Management Systems, AI & ML